

	External Assessment Specifications Document			
Curriculum Code	Qualification Title		NQF Level	
252301001	Occupational Certificate: Cloud Administrator		4	
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DQP Representative Signature

Date

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Date

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1 ASSESSMENT STRATEGY

1.1 Assessment Model

An external integrated summative assessment, conducted through the relevant QCTO Assessment Quality partner is required for the issuing of this qualification. The external integrated summative assessment will focus on the exit level outcomes and associated assessment criteria.

The assessment will be conducted through written assessment and the evaluation of practical tasks at decentralised approved assessment sites by a panel of assessors evaluated by assessor(s) registered with the AQP within a period of 1 day.

1.2 Qualification Purpose

The purpose of this qualification is to prepare a learner to operate as a Cloud Administrator

A Cloud Administrator implement cost-effective cloud-based systems and the cloud management service and transition locally hosted infrastructure to the cloud. They monitor, maintain and troubleshoot networks of cloud platforms and computing resources ensuring the seamless delivery of all cloud services and maintain protocols to secure cloud environments against unauthorised access, threats and other risks.

A qualified learner will be able to:

- Implement the cloud management service.
- Monitor and maintain specific elements of the cloud, cloud users, the cloud environment and overall cloud performance.
- Maintain security protocols to secure cloud environments against unauthorised access, threats and other risks.

1.3 Assessment Standards

1.3.1 Assessment Standards for the Qualification

Integrated Assessment Focus Area 1

Implement cloud solutions and services related to cloud computing, networks, security, and storage in a cloud environment (20%)

Associated Assessment Criteria

- An overall understanding of the cloud, cloud infrastructure and cloud related services are demonstrated
- Public or private cloud systems and appropriate network configurations are set up
- Connecting, routing and networking are implemented to integrate systems for hosting the solution on the cloud
- Systems infrastructure for new and existing environments is documented
- Appropriate testing techniques are executed when deploying cloud services

Integrated Assessment Focus Area 2

Monitor and maintain servers and networks for optimal cloud performance and seamless delivery of all cloud services (30%)

Associated Assessment Criteria

- Specific elements of the cloud, cloud users, the cloud environment and overall cloud performance are monitored and maintained

- The capabilities and performance of the systems of the cloud ecosystem are monitored
- Networks of cloud platforms and other computing resources are tested, diagnosed and remediated
- Maintenance duties, such as responding to trouble tickets and alerts, developing patches or fixes are performed for any issues that arise and operational problems are resolved
- Workload is balanced and deployed in an automated way

Integrated Assessment Focus Area 3

Monitor on-demand availability of data, data storage and storage capacity in line with the life cycle of data storage and hardware architecture (30%)

Associated Assessment Criteria

- On-demand availability of data is monitored and maintained
- Data storage solutions and storage capacity are monitored and maintained and long-term archival storage is configured
- Regular back-up of data is performed and readiness for restoration when needed is ensured
- Encryption is applied to protect data in rest, use and transit states

Integrated Assessment Focus Area 4

Apply security configurations and compliance controls to meet cloud infrastructure requirements (20%)

Associated Assessment Criteria

- Cloud environments, including applications, data, and information are secured against unauthorized use/access, distributed denial of service (DDOS) attacks, hackers, malware, and other risks
- The appropriate access control lists (ACL) are applied to the target objects to meet access requirements.
- Security testing techniques are applied
- In the case of breach in security, appropriate disaster recovery methods and appropriate steps to ensure business continuity are applied
- Systems standards as they relate to hardening and security within the cloud infrastructure are maintained

1.3.2 Assessment Standards for Phases

None

1.3.3 Assessment Standards for Part Qualifications

None

2 ASSIGNMENTS TO BE EVALUATED EXTERNALLY

None

3 CRITICAL ASPECTS OF THE INTERNAL ASSESSMENTS TO BE ASSESSED EXTERNALLY

- None

4 CRITERIA FOR THE REGISTRATION OF ASSESSORS

Qualifications:

- Assessors and moderators: accredited by the MICT SETA
- Subject matter experts: NQF 5 in any industry relevant qualification

Experience:

- 1 year's experience in the IT industry
- 1 year's experience in assessment and moderation

5 FOUNDATIONAL LEARNING

Required

6 ELIGIBILITY REQUIREMENTS FOR THE EXTERNAL ASSESSMENT

In order to qualify for the external summative assessment learners must provide proof of completion of all required modules by means of a statement of results and statement of work experience as well as proof of successful completion of the following subjects and modules or alternative programs where applicable, including Foundational Learning Competence.

6.1 Qualification

	<i>Proof of Knowledge Modules</i>			<i>OR</i>			<i>Proof of Alternative Programmes</i>	
<i>Number</i>	<i>Title</i>	<i>NQF Level</i>	<i>Credits</i>		<i>Number</i>	<i>Title</i>	<i>NQF Level</i>	<i>Credits</i>
252301-001-00-KM-01	Introduction to Cloud Computing	4	8					
252301-001-00-KM-02	Cloud Computing Elements	4	8					
252301-001-00-KM-03	Containers, Cloud Native and Kubernetes	4	4					
252301-001-00-KM-04	Cloud Security, Risks, Vulnerabilities and Mitigation	4	3					
252301-001-00-KM-05	Data and Databases in the Cloud	4	6					
252301-001-00-KM-06	Fundamentals of Cloud Computing	4	3					

	Platforms							
252301-001-00-KM-07	Introduction to Cloud Computing Governance, Legislation and Ethics	4	1					
252301-001-00-KM-08	4IR and Future Skills	4	4					
252301-001-00-KM-09	Design Thinking Principles for Innovation	4	1					

	<i>Proof of Practical Skill Modules</i>			<i>OR</i>			<i>Proof of Alternative Programmes</i>	
<i>Number</i>	<i>Title</i>	<i>NQF Level</i>	<i>Credits</i>		<i>Number</i>	<i>Title</i>	<i>NQF Level</i>	<i>Credits</i>
252301-001-00-PM-01	Conduct Effective Server Maintenance and Optimisation	4	8					
252301-001-00-PM-02	Monitor On-Demand Availability of Data Storage, Storage Capacity and Storage Services over the Internet	4	8					
252301-001-00-PM-03	Provide Cloud Network Administration and Support for Optimal Performance	4	16					

252301-001-00-PM-04	Ensure Integrity of the Cloud Ecosystem by Implementing Security Measures	4	8					
252301-001-00-PM-05	Access and Visualise Structured Data Using Spreadsheets	4	8					
252301-001-00-PM-06	Participate in a Design Thinking for Innovation Workshop	4	3					
252301-001-00-PM-07	Function Ethically and Effectively in the Workplace	4	4					

And

Statement of Work experience

6.2 Phase 1

No Phase Tests applicable

7 PART QUALIFICATIONS

None